Causal Structure: Load, Capacity, and Boundary in Dimensional Human Field Theory (DHFT)

DHFT Technical Note: Causal Structure

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April 14, 2026 — Version 1
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Abstract

This document defines the causal structure governing system behavior in Dimensional Human Field Theory (DHFT). The system is specified over invariant variables Load (*L*), Capacity (*C*), and Boundary (*B*).

The relations defined here establish directional dependencies between variables and constrain how system state changes occur under load. Causality is treated as structural and non-interpretive.

No additional variables are introduced. All causal relations remain constrained by the minimal system defined in DHFT.

I. Scope

This document defines the directional relations governing system behavior in DHFT. It specifies how load, capacity, and boundary conditions interact to produce changes in system state.

No interpretive, psychological, or domain-specific constructs are introduced.

II. Variable Reference

The system is defined over three invariant variables: Load (*L*), Capacity (*C*), and Boundary (*B*). These variables are fixed at the minimal layer and are not expanded or redefined within this document.

III. Causal Relations

Boundary conditions govern load entry, containment, and transfer. Load acts on the system and interacts with capacity to determine system state.

Capacity constrains the system's ability to process load and does not regulate load entry.

System state is determined by load relative to capacity under boundary conditions.

IV. Directional Dependency

Causal dependency proceeds from boundary conditions to load, and from load to capacity constraint.

Boundary conditions determine load exposure. Load determines system state relative to capacity. Capacity limits the system's ability to maintain structural continuity under load.

No valid relation permits capacity to regulate boundary conditions or load entry at the minimal layer.

V. Causal Constraint

All valid descriptions of system behavior must preserve the directional relations defined here. Reversal or reinterpretation of causal order is not permitted at the minimal layer.

Causal relations are structural and do not depend on perception, interpretation, or intent.

VI. Statement

The causal structure defined here establishes directional dependencies governing system behavior in DHFT. All valid system descriptions at the minimal layer preserve these relations.

VII. System Reference

The system defined in this document is specified in *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*, DOI: 10.5281/zenodo.19075948.

All valid system descriptions reduce to Load, Capacity, and Boundary.

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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